

Parallel framework of a multi-graph convolutional network and gated recurrent unit for spatial–temporal metro passenger flow prediction

Shuguang Zhan^a, Yi Cai^b, Cong Xiu^{b,*}, Dajie Zuo^b, Dian Wang^a, Sze Chun Wong^c

^a School of Automotive and Transportation Engineering, Hefei University of Technology, Hefei 230009, China

^b School of Transportation and Logistics, Southwest Jiaotong University, Chengdu 610031, China

^c Department of Civil Engineering, The University of Hong Kong, Hong Kong, China

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ABSTRACT

Metro passenger flow prediction is a critical problem in metro transport systems. However, recent studies have either overlooked spatial information on the metro network or primarily focused on modeling spatial dependencies using only the physical topology. To achieve accurate metro passenger flow (inflow and outflow at each station of a network) prediction, this study proposes a joint prediction model that combines the multi-graph convolution network and the gated recurrent unit (GRU). In addition to exploring location topology relationships, the proposed model selects two non-Euclidean spatial dependencies in metro passenger flow prediction to design essential graph elements as part of the stacked spatial block. Three spatial relationships (adjacency, similarity, and correlation) are integrated in parallel with the GRU network. The metro passenger flow prediction framework ASC-GRU (adjacency, similarity, correlation, and gated recurrent unit) is designed to mitigate the distortion of results during the capturing of passenger flow spatial–temporal features. Finally, ASC-GRU is tested using two datasets from the Hangzhou and Shanghai metro networks in China, and the error metrics of different models are compared and analyzed to verify the effectiveness and feasibility of ASC-GRU. The test results demonstrate that the proposed model outperforms other baseline models in passenger flow prediction over long time intervals and large networks. In particular, compared with the best performance of the baselines, the average reduction is around 3%, 12% and 13% in metrics of MAPE, MAE and RMSE, respectively.

1. Introduction

Metros play an essential role in urban transit systems owing to their convenience, high efficiency, and economy. For example, in 2018, the passenger flow of the Beijing metro reached 10 million per day (Tang et al., 2019). Such a huge passenger flow may cause frequent congestion and resource waste. The accurate prediction of passenger flow (i.e., inflow and outflow of each station on a network, also called intra-station passenger flow in this study) is necessary to improve transportation efficiency and the service level, and it has become an important research area in intelligent transportation systems (Veres & Moussa, 2019; Lee et al., 2021). Accurate passenger flow prediction is vital to improve metro usage efficiency, route optimization, and traffic scheduling. Moreover, owing to the close connection and complex spatial dependence between stations in the metro network, passenger flow characteristics vary over time, making it challenging to capture the

spatial–temporal features and predict passenger flow accurately.

Studies on metro passenger flow forecasting have widely adopted classical mathematical models such as the auto-regressive integrated moving average (ARIMA) (Feng et al., 2016; Yan et al., 2018) and the Kalman filter (Kumar, 2017), as well as deep learning models such as recurrent neural networks (RNNs) (Cheng et al., 2012; Ma et al., 2015; Fu et al., 2016) and convolutional neural networks (CNNs) (Chen et al., 2018; Ma et al., 2018; Bai et al., 2018; Lv et al., 2020). Despite the breakthroughs in research related to deep learning methods in the transportation field, there are still some challenges in achieving more effective and accurate metro passenger flow prediction.

First, numerous traditional metro passenger flow prediction models treat the prediction task as a standard time-series prediction problem. However, they often overlook the spatial dependence between stations (Fu et al., 2016; Cui et al., 2018). Second, the models adopted in most recent studies only learned the local spatial dependency of neighboring

* Corresponding author.

E-mail addresses: shuguangzhan@hfut.edu.cn (S. Zhan), cc9924@my.swjtu.edu.cn (Y. Cai), xiucong001@my.swjtu.edu.cn (C. Xiu), zuodajie@swjtu.cn (D. Zuo), dianwang@hfut.edu.cn (D. Wang), hhecwsc@hku.hk (S. Chun Wong).

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stations and could not fully capture the inter-station flow patterns (i.e., flow patterns between non-adjacent stations) in a metro system. Some studies (Huang et al., 2014; Cui et al., 2019) directly built adjacency graphs based on the physical topologies of the traffic systems. However, although building adjacency graphs based on physical topologies is straightforward, it is suboptimal for metro passenger flow prediction, as only the local spatial dependency of neighboring stations is considered. The topology of a metro system is an irregular graph, and the data structures of the system are non-Euclidean. For example, even if two metro stations are not directly linked in the real-world metro network, the ridership of these two stations may exhibit a strong correlation if their locations share the same functionality (e.g., commerce or residential). Therefore, constructing reasonable graphs for comprehensive spatial features in metro passenger flow prediction is important.

Given the aforementioned problems, two types of spatial dependencies with professional domain knowledge are considered in our study, passenger flow similarity and passenger flow correlation, in addition to the station adjacency graph. Each area in a city has a specific role, such as work or entertainment. Two metro stations in different regions may exhibit similar passenger flow evolution patterns if their locations share the same functionality. Therefore, considering passenger flow similarity is essential. Furthermore, the passenger flow from the start point to the end point is directional, and the passenger flow direction represents the correlation between the two stations. The existence of large passenger flow records between two stations indicates a frequent interaction between them. To account for this interaction, passenger flow correlation needs to be considered in metro passenger flow prediction.

In this paper, a parallel-structure deep learning model ASC-GRU (adjacency, similarity, correlation, and gated recurrent unit), which comprises a multi-graph convolutional network and a gated recurrent unit (GRU), is proposed to predict intra-station passenger flow. First, the spatial dependencies of station adjacency, passenger flow similarity, and passenger flow correlation in a metro network are proposed. Then, three graphs (station adjacency graph, passenger flow similarity graph, and passenger flow correlation graph) (Liu et al., 2020) are selected to compose the stacked spatial block. The stacked spatial block is integrated with the GRU time-series forecasting module in parallel form to build a joint forecasting model. This model incorporates the spatial capture power of the multi-graph convolutional network and the temporal capture power of the GRU network to fully explore the ridership evolution patterns. Finally, real datasets from the complex metro networks of Hangzhou and Shanghai in China are experimentally evaluated, and the results show that our approach outperforms existing methods in various comparison circumstances.

This study makes three main contributions:

- A new parallel deep learning architecture ASC-GRU, composed of two modules considering both spatial and temporal dependencies for metro passenger flow prediction, is introduced. Specifically, ASC-GRU consists of a multi-graph convolutional network and a GRU.
- The ASC-GRU architecture achieves good performance by combining the three graphs (station adjacency graph, passenger flow similarity graph, and passenger flow correlation graph) in the multi-graph convolution module.
- Extensive experiments on two metro passenger flow datasets show that the proposed ASC-GRU outperforms baseline methods for passenger flow prediction, and the ASC-GRU model is particularly beneficial for long-term forecasting in a highly complex metro network.

The rest of this paper is organized as follows. Section 2 reviews related studies. Section 3 defines the key to modelling metro networks and details the research questions. Section 4 elucidates the proposed ASC-GRU model. Section 5 describes the dataset setup and specific implementation procedures and presents an analysis of the experimental

results. Finally, Section 6 provides conclusions.

2. Literature review

This study uses graph convolutional networks (GCNs) for metro passenger flow prediction. Therefore, the relevant literature is categorized into two aspects: metro passenger flow prediction and GCNs.

2.1. Metro passenger flow prediction

In recent years, traffic state prediction, particularly in the context of metro passenger flow, has become a prominent research area in the transportation field (Zhang et al., 2011; Tedjopurnomo et al., 2020), playing a crucial role in helping metro companies make informed decisions and ensures the safety of passenger travel (Ye et al., 2020; Zhang et al., 2020). With the advancements in data acquisition technology and the availability of traffic-related big data, data-driven methods have gained popularity (Nagy and Simon, 2018; Tedjopurnomo et al., 2020). Unlike model-based methods, such as the cell transmission model (Wei et al., 2013) and store-and-forward model (Aboudolas et al., 2009), data-driven methods leverage statistical regularities and distributions in historical data to infer changes in traffic states (Jiang et al., 2022; Shahriari et al., 2020). Existing data-driven methods fall into two categories: parametric-based models and non-parametric models (Nagy and Simon, 2018; Zhang et al., 2011). Parametric models employ regression functions to forecast traffic based on historical data, such as the autoregressive integrated moving average (ARIMA), e.g., Gu et al. (2011), and its variants, e.g., the Kohonen ARIMA (Lee and Fambro, 1999), ARIMAX (Williams, 2001), and seasonal ARIMA (Williams and Hoel, 2003). Although these models are user-friendly, their predictive performance is limited due to their inability to handle nonlinearity and non-stationarity in traffic data. Non-parametric models have emerged as a solution to this problem (Shi et al., 2020; Zhang et al., 2011). Machine learning-based approaches are the most representative, including K-nearest neighbor (KNN), support vector machine (SVM) and Kalman filter method. With sufficient historical data, KNN (Arroyo and Maté, 2009; Yu et al., 2019), SVM (Castro-Neto et al., 2009; Wu et al., 2004) and Kalman filter models (Kumar, 2017) can learn statistical patterns and achieve more accurate predictions. However, these mathematical models are based on a linear theory, ignoring the nonlinearity and dynamics of the original metro passenger flow data, and thus cannot meet the real-time requirement in prediction. Deep learning addresses these limitations of traditional mathematical models by constructing various network structures, facilitating the extraction of underlying features from passenger flow data, and accommodating dynamic temporal changes.

Several studies have achieved innovative results by applying deep learning models (Polson & Sokolov, 2017; Yu et al., 2017; Yao et al., 2018; Do et al., 2019; Liu et al., 2019; Qu et al., 2019; Jin et al., 2021). Among the deep learning models, RNNs were initially widely adopted for time series prediction owing to their ability to process inputs of arbitrary length using internal memory (Cheng et al., 2012). However, the RNN model faces challenges in handling long sequences owing to gradient disappearance. Consequently, its variants, long-short-term memory (LSTM) (Ma et al., 2015; Cui et al., 2018) and GRU (Fu et al., 2016), have been used to overcome the gradient disappearance issue. The above works are widely used to capture the time-varying characteristics of passenger flow at the station level. However, the station-level prediction methods that solely focus on temporal dependence while neglecting spatial dependence still have room for improvement in terms of prediction accuracy.

To address the limitations of traditional station-level prediction methods, many researchers have focused on network-scale passenger flow prediction models that combines spatial dependencies and temporal features (Wu et al., 2020; Guo et al., 2021). However, the inherent topological irregularities and non-Euclidean data in these models restrict their ability to accurately represent connections between two

stations in a metro network. These models primarily emphasize capturing temporal components quickly, which hampers the accurate prediction of metro passenger flow as studying the spatial characteristics of passenger flow becomes challenging.

A growing number of scholars have recognized that the spatial structure of a transportation network leads to interactions between the passenger flows of adjacent stations. CNNs have been used to convert real-world metro networks into images to incorporate spatial features in passenger flow prediction (Bai et al., 2018; Chen et al., 2018; Shen et al., 2018; Wu et al., 2018; Lv et al., 2020; Shin & Yoon, 2020). CNNs can represent Euclidean features by calculating the distance between stations to capture spatial dependencies. However, CNN-based models do not fully consider the non-Euclidean structural features present in metro networks, resulting in incomplete learning of connectivity. Additionally, the real-world network structure may not always align with the spatial relationships constructed by CNN.

In contrast, GCNs offer a novel approach to traffic data modeling, enabling prediction with complete extraction of spatial features while compensating for the shortcomings of CNN models in portraying factual topological information in traffic networks (Lin et al., 2013; Wu et al., 2018; Zhu et al., 2022; Ren et al., 2023). Li et al. (2017) proposed a diffusion convolutional recurrent neural network (DCRNN) that combines GCN and GRU in a Seq2Seq framework to predict traffic flow. A similar approach, T-GCN, proposed by Zhao et al. (2019), has gained significant attention. Zhang et al. (2019) aimed to improve model input quality by considering multiple factors but lacked adequate selection of spatial-temporal features and failed to establish joint spatial-temporal prediction models. Cui et al. (2019), Jin et al. (2020), and Ma et al. (2022) proposed integrating convolutional modules with LSTM and GRU through a sequential structure to capture both spatial and temporal features for prediction.

Several studies introduce attention mechanisms, to effectively build dependencies in sequences for traffic prediction models, such as the attention-based spatial-temporal graph convolutional network (ASTGCN), graph multi-attention network (GMAN) and multivariate time-series-based graph neural network (MTGNN) (Guo et al., 2019; Wu et al., 2020; Zheng et al., 2020). However, capturing spatial relations for different time slots with a shared graph convolutional kernel in the above works may not be efficient enough for dynamically changing traffic patterns (James, 2022; Lee and Rhee, 2022).

Based on multi-graph, various approaches have been proposed for modeling the traffic state, such as OD ride-sourcing demand (Ke et al., 2021) and network-level traffic speed (Zhang et al., 2021). Notably, unlike road, metro operation regularities and passenger travel behaviors exhibit a typical tunnel effect. This phenomenon involves passengers' preference for using the metro for long-distance travel rather than opting for nearby stations, posing a challenge for physical metro networks to accurately capture this behavior. Liu et al. (2020) proposed the physical-virtual collaboration graph network (PVCN), which constructs virtual topological graphs based on the similarity of traffic flow between stations in the metro system. Chen et al. (2021) employed a stacked LSTM and GCN module, integrating a K-hop matrix to represent connections, distances, and passenger flow correlations.

Although the methods above capture various spatial relationships, the constructed spatial correlations are primarily operated by stacked GCN operations. However, few studies have explored how to leverage these constructed graphs, such as in parallel or stacked fashion. Additionally, it is noteworthy that for practical applications of metro passenger flow forecasting, existing GCN models often underutilize passenger flow spatial-temporal information. In contrast, this research employs more reasonable graphs (station adjacency graph, passenger flow similarity graph, and passenger flow correlation graph) to depict inter-station dependencies in the metro system. Furthermore, a stacked structure comprising these three graphs is integrated with the GRU time-series forecasting module in a parallel manner, enhancing the temporal and spatial modules' forecasting capabilities. This approach offers a

more comprehensive and robust solution for metro passenger flow prediction.

2.2. GCNs

Metro passenger flow data encompass both Euclidean structure and non-Euclidean characteristics. Although CNN/RNN can successfully handle Euclidean data, they face difficulty in dealing with non-Euclidean data. GCNs have been proposed to automatically learn feature representation on graphs to address the non-Euclidean issue. Bruna et al. (2013) first proposed a GCN by combining spectral graph theory and deep neural networks. Defferrard et al. (2016) improved GCN by simplifying the estimate using ChebNet inequality to realize a more competitive model performance. Kipf et al. (2016) introduced 1stChebNet as an extension of the ChebNet inequality to reduce computational costs. The edge weights in different traffic networks play a decisive role in graph convolution models, as the prediction depends on the graph weight matrix. In early studies, only Euclidean distance was used as the graph element to preserve the physical topological information in the real world and extract spatial features (Huang et al., 2014; Cui et al., 2019; Jin et al., 2020). However, traffic data contain latent non-Euclidean spatial dependencies, such as highly correlated traffic patterns between sites that are not directly adjacent to each other in the transport network. Recent studies have incorporated other virtual graphs based on prior knowledge to improve the effectiveness of GCN. For instance, Zhang et al. (2022) utilized regional functional similarity graphs and public transportation connectivity graphs, in addition to the basic regional adjacency graph, for online car demand prediction. Chai et al. (2018) used three graphs (distance graph, interaction graph, and correlation graph) to predict shared bike traffic, considering the number of rides between two stations and the inter-station traffic correlation (calculated using the Pearson coefficient) and using the inverse of the distance as edge weights.

The abovementioned studies primarily focused on implementing traffic forecasting on road networks. However, metro networks feature irregular physical topologies and different non-Euclidean data structures from road networks. For instance, Lee et al. (2021) considered different speed patterns and driving directions of vehicles as non-Euclidean dependencies for road traffic prediction. In contrast, metro passenger flow prediction does not consider these two spatial features owing to the uniform travel pattern of the metro system. Furthermore, for road networks, the shortest interlink distance between any two sites is often selected as one of the graph elements in regular physical topologies (Do et al., 2019; Jin et al., 2020). In contrast, metro systems typically utilize adjacency relationships to denote the connectivity between stations in irregular physical topologies. Therefore, the graph elements constructed for road traffic forecasting may not be directly applicable to metro systems.

Liu et al. (2020) proposed three graphs—the station adjacency graph, passenger flow similarity graph, and passenger flow correlation graph—to fully explore the relationship between two metro stations, and they selected a parallel spatial block consisting of these three graphs to capture the spatial features of metro passenger flow. Although these three graphs are indeed valuable, combining them in a parallel manner may not be the most efficient approach to fully capture the spatial characteristics. Instead, combining the graphs in a stacked manner is more efficient in capturing spatial characteristics, as demonstrated by Lee et al. (2022) for traffic speed forecasting. Therefore, in the present study, a stacked structure composed of the three graphs (station adjacency graph, passenger flow similarity graph, and passenger flow correlation graph) is built as spatial modules in metro passenger flow prediction. Liu et al. (2020) embedded the spatial block, mainly consisting of the three graphs, into a GRU temporal module for spatial-temporal dependency learning in metro passenger flow prediction. However, Ma et al. (2018) discovered that combining the spatial module and the temporal module in a parallel manner significantly improved

the accuracy of metro ridership prediction, as compared with directly embedding the spatial module into the temporal module. On the basis of these findings, this study integrates the spatial module with the GRU temporal module in a parallel manner to enhance prediction accuracy in metro passenger flow forecasting. This approach is expected to provide better results by effectively capturing both spatial and temporal dependencies in the prediction process.

3. Definition and problem formulation

In this section, the critical concepts of metro network modeling are defined, and the problem is described.

- Definition of metro network graph:** The metro network is depicted with a weighted directed graph $G = (V, E, W)$, which encompasses stations, connecting edges, and associated edge weights. V denotes the set of stations in the metro network, $|V| = N$, and N is the total number of stations in the metro network. E denotes the set of edges connecting stations, and it encompasses all of the edges connecting any two stations. $W \in R^{N \times N}$ denotes the edge weights, and $W_{(ij)}$ denotes the weight of edge $(i, j) \in E$ connecting stations $i \in V$ and $j \in V$.

Three types of graphs, i.e., the station adjacency graph $G_A = (V, E_A, W_A)$, the passenger flow similarity graph $G_S = (V, E_S, W_S)$, and the passenger flow correlation graph $G_C = (V, E_C, W_C)$, are introduced. The three graphs feature the same sets of stations V , indicating that all three graphs share all of the stations in the metro network. However, the three graphs have different sets of edges and edge weights. E_A , E_S , and E_C denote different sets of edges owned by the three graphs. In a particular metro network graph $G_P = (V, E_P, W_P)$ ($P = A, S, C$), and $W_{P(ij)}$ denotes the weight of the edge $(i, j) \in E_P$ connecting station $i \in V$ and station $j \in V$ in graph G_P . The three graphs are detailed in section 4.2.1.

- Description of the metro passenger flow prediction problem:** Metro passenger flow prediction can be formulated as a function mapping process, denoted as f , as depicted in equation (1). X_t^i denotes the metro passenger flow of station i in time interval t ($X_t^i \in R^2$ denotes the number of passengers entering (inflow) or exiting (outflow) station i during time interval t). $X_t = (X_t^1, X_t^2, \dots, X_t^N)$ indicates passenger flows in all stations of the metro network ($X_t \in R^{2 \times N}$, N is the number of stations). The historical metro passenger data are aggregated using specific time intervals to form n time series, which is mapped by the function f to predict the passenger flow for the future m time series.

$$[X_{t-n+1}, X_{t-n+2}, \dots, X_t] \xrightarrow{f} [X_{t+1}, X_{t+2}, \dots, X_{t+m}] \quad (1)$$

4. Methodology

This section presents the framework of the proposed model (ASC-GRU) for metro passenger flow prediction. The section includes the definition of spatial graph elements, the construction of spatial blocks (multi-graph convolutional networks), and the construction of time blocks (GRU).

4.1. General framework

As depicted in Fig. 1, the proposed parallel deep learning model ASC-GRU comprises a multi-graph convolutional network (spatial block) and a gated recurrent neural network (temporal block). First, the station adjacency graph, passenger flow similarity graph, and passenger flow correlation graph are set. A combination structure of the three graph elements is used to represent the complex spatial relationship in the metro network. Then, the GRU is applied to capture the temporal characteristics of metro passenger flow. Finally, the GCN and GRU results are integrated to complement each other, and a parallel

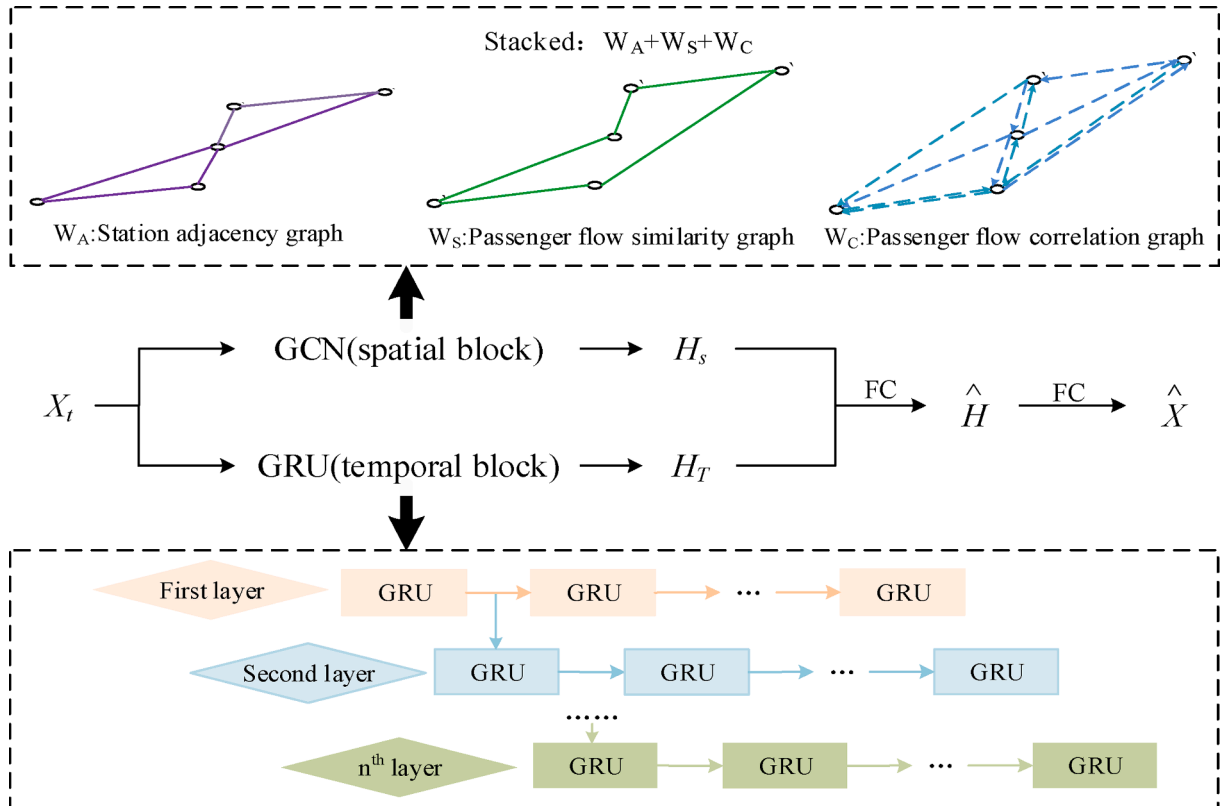


Fig. 1. ASC-GRU consisting of a multi-graph convolutional network and a GRU.

combination is built to achieve a more accurate prediction of metro passenger flow. The steps to build and combine the GCN and GRU modules are detailed below. Further details on the general framework are given in Section 4.4.

4.2. Building spatial blocks

Before the GCN to capture spatial dependencies is described, the convolution expression derived from 1stChebNet (Kipf & Welling, 2016) is first presented. The learning parameter of the GCN is expressed by θ , and the output feature $f(X_t^i)$ is formulated as equation (2).

$$f(X_t^i) = \theta_i X_t^i + \sum_{j \in H_i} W_{P(i,j)} \odot \theta_P X_t^j \quad (2)$$

Here, $\theta_i X_t^i$ is the self-loop of all selected graphs; that is, the selected graph updates itself with metro passenger flow data at station i , where θ_i is the learning parameter for station i . H_i represents the neighbor set of station i , and $j \in H_i$ means that j belongs to the neighborhood of i (i.e., stations i and j have spatial relationships (adjacency, similarity, or correlation), and $W_{P(i,j)} \neq 0$). The situation in which station i is affected by some other stations that are highly correlated with it (i.e., stations in set H_i) is fully considered. P denotes the selected graph $G_P = (V, E_P, W_P)$ (details can be found in the definition of metro network graphs in Section 3). $W_{P(i,j)}$ denotes the weight of edge $(i,j) \in E_P$ connecting station $i \in V$ and station $j \in V$ in graph P . $\odot$ denotes the dot product operation, and θ_P is the learning parameter of the selected graph (i.e., graph P).

4.2.1. Definition of spatial graph elements

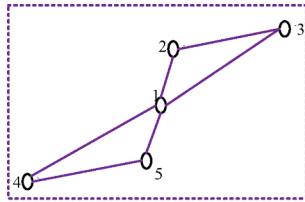
Graph definition is a key step for GCN. If the defined graphs cannot effectively encode the correlations between the passenger flows of two

stations, it would not help the model learning and might even degrade the prediction performance (Lv et al., 2020). GCN utilizes the Laplacian matrix, and various studies have used a binary-encoded weight matrix to depict the connectivity between two stations. If station i is connected to station j , the weight in the matrix is $A_{(ij)} = 1$; otherwise, $A_{(ij)} = 0$. However, as mentioned above, the topologies of a metro network are irregular graphs, and the structures of metro passenger flow data are non-Euclidean. Therefore, using only spatial distances to compose a simple binary weight matrix between stations in a metro system has application limits. To this end, the construction of a multi-graph convolution is suggested to effectively capture the spatial features of passenger flow. The construction process of the three spatial graphs (station adjacency graph, passenger flow similarity graph, and passenger flow correlation graph) is described below (Liu et al., 2020).

Station adjacency graph: The connection between two metro stations affects the passenger flow relationship between the two stations, and two stations alongside each other are more strongly related. In the real world, when two stations i and j are adjacent, an edge (i,j) in the edge set E_A connecting them $((i,j) \in E_A)$ defines a physical connection matrix $A \in R^{N \times N}$. The value of $A_{(ij)}$ is determined by whether there is a metro line connecting adjacent stations i and j , and it can be calculated using equation (3).

$$A_{(ij)} = \begin{cases} 1, & \text{if } (i,j) \in E_A \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Equation (3) indicates that if there exists a metro line connecting adjacent stations i and j , $A_{(ij)} = 1$; otherwise, $A_{(ij)} = 0$. Fig. 2(b) shows the station adjacency score matrix based on the connection relationship of metro stations in Fig. 2(a) (for example, station 2 is directly connected



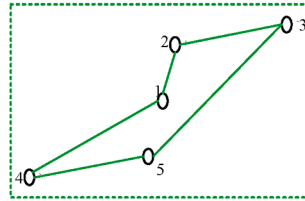
(a) Adjacent graph

0	1	1	1	1
1	0	1	0	0
1	1	0	0	0
1	0	0	0	1
1	0	0	1	0

Normalization

0	0.25	0.25	0.25	0.25
0.5	0	0.5	0	0
0.5	0.5	0	0	0
0.5	0	0	0	0.5
0.5	0	0	0.5	0

(c) Adjacent edge weight



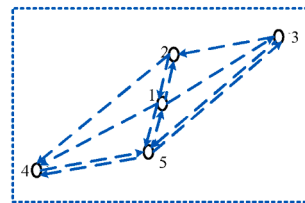
(d) Similarity graph

0	0.73	0.05	0.56	0.06
0.73	0	0.47	0.07	0.06
0.05	0.47	0	0.04	0.51
0.56	0.07	0.04	0	0.42
0.06	0.06	0.51	0.42	0

Selection
Normalization

0	0.57	0	0.43	0
0.61	0	0.39	0	0
0	0.48	0	0	0.52
0.57	0	0	0	0.43
0	0	0.55	0.45	0

(f) Similarity edge weight



(g) Correlation graph

0.01	0.58	0.04	0.03	0.14
0.50	0.05	0.40	0.03	0.02
0.39	0.03	0.01	0.03	0.54
0.35	0.25	0.01	0.05	0.34
0.40	0.04	0.29	0.26	0.01

Selection
Normalization

0	0.63	0	0	0.37
0.56	0	0.44	0	0
0.42	0	0	0	0.58
0.37	0.27	0	0	0.36
0.42	0	0.31	0.27	0

(i) Correlation edge weight

(h) Correlation score matrix

Fig. 2. Weight matrix of the three graphs used by the GCN module.

to stations 1 and 3, $A_{(2,1)} = 1$, and $A_{(2,3)} = 1$, and the rest values in the second row are 0). The elements on the diagonal (where the starting and ending points of the edges are the same station) are set to 0 to avoid repeated calculation of the self-loop, and the update of the self-loop is computed only once in a unified manner in equation (2).

Finally, each row in the adjacency score matrix (Fig. 2(b)) is linearly normalized, and a new weight $W_{A(i,j)}$ for edge (i,j) connecting station i and station j in graph G_A is obtained. The weight is computed using equation (4), and Fig. 2(c) shows the updated weight matrix.

$$W_{A(i,j)} = \frac{A_{(i,j)}}{\sum_{k=1}^N A_{(i,k)}} \quad (4)$$

Passenger flow similarity graph: Each area in a city has a distinct function, such as commerce (shopping centers, markets), residential (apartments, dormitories), and leisure (cinemas, bars). Metro stations located in areas with similar roles may feature similar passenger flow patterns.

According to the above observation, identifying the function of each area in a city according to point of interest (POI) mining analysis is necessary. First, the historical POI data of a city and the classification accuracy are fed into the neural network for calculation. Then, the spatial-temporal clustering results are obtained, and the dominant attributes of the landmarks are analyzed.

The passenger flow similarity graph is proposed to explore the degree of similarity of passenger flow patterns between two metro stations located in areas with similar roles. First, a similarity score matrix $P \in R^{N \times N}$ (Fig. 2(e)) based on the station passenger flow similarity relationship graph (Fig. 2(d)) is constructed. The similarity value $P_{(i,j)}$ is the historical passenger flow data between two stations i and j . $P_{(i,j)}$ is computed using the dynamic time warping (DTW) algorithm (Berndt & Clifford, 1994) according to metro passenger flows X^i and X^j of stations i and j . In experiments, the similarity score matrix P is calculated using the experimental dataset collected for prediction in the metro systems. The calculation process is shown by equation (5).

$$P_{(i,j)} = \exp(-DTW(X^i, X^j)) \quad (5)$$

The DTW algorithm is implemented using Python, and it selects the optimal path between two temporal sequences X^i and X^j according to the principle of minimum distance to exactly measure the similarity value. A smaller DTW value means a larger $P_{(i,j)}$ value. The relationship between stations i and j is stronger if the passenger flow patterns between them feature a higher similarity value. As for the passenger flow historical data of the same station, the similarity value $DTW(X^i, X^i)$ tends to be infinitely large. Therefore, the value of $P_{(i,i)}$ on the diagonal of the similarity matrix, calculated by equation (5), tends to be infinitely small and is thus set to 0.

According to the similarity score matrix (Fig. 2(e)), the station pairs with higher similarity values are selected (i.e., the black data, corresponding to larger similarity weights, are selected, while the gray data, corresponding to smaller weights, are discarded) to build the edge set similarity values that are normalized with formula (6), and new edge weights $W_{S(i,j)}$ are obtained. The updated similarity value matrix is shown in Fig. 2(f).

$$W_{S(i,j)} = \frac{P_{(i,j)}}{\sum_{k=1}^N P_{(i,k)} \cdot L_{(i,k,E_S)}} \quad (6)$$

In equation (6), if there exists an edge (i,k) connecting stations i and k in the edge set E_S , $L_{(i,k,E_S)} = 1$; otherwise, $L_{(i,k,E_S)} = 0$.

Passenger flow correlation graph: The passenger riding behavior from the origin to the destination is directional, and the direction of passenger riding flow represents the correlation between two stations. The existence of more passenger flow records between two stations indicates a more frequent interaction between them. Therefore, stations with large passenger flow records are connected to learn the ridership

interaction between them.

According to the above observation, the passenger flow correlation graph is proposed, and a correlation score matrix $C \in R^{N \times N}$ (Fig. 2(h)) based on the passenger flow correlation graph (Fig. 2(g)) is constructed. The value of $C_{(i,j)}$ in the matrix is calculated using equation (7).

$$C_{(i,j)} = \frac{O_{(i,j)}}{\sum_{k=1}^N O_{(i,k)}} \quad (7)$$

Here, $O_{(i,j)}$ represents the total number of passengers that travel from station j to station i . Because the number of passengers that both enter and exit from the same station i is small (it rarely happens in practical scenarios), the magnitude of the element on the diagonal is usually tiny; that is, the $C_{(i,i)}$ value is small.

Finally, pairs of stations with large correlation values are selected to build the edge set. The size of the weight value $W_{C(i,j)}$ is updated via normalization calculation using equation (8). If there exists an edge (i,k) connecting stations i and k in the edge set E_C , $L_{(i,k,E_C)} = 1$; otherwise, $L_{(i,k,E_C)} = 0$. The new edge weight matrix is shown in Fig. 2(i).

$$W_{C(i,j)} = \frac{C_{(i,j)}}{\sum_{k=1}^N C_{(i,k)} \cdot L_{(i,k,E_C)}} \quad (8)$$

4.2.2. Multi-graph convolutional networks

With the three graph elements W_A , W_S and W_C defined above, the spatial block is constructed using multi-graph convolutional networks to capture the complex spatial dependencies between the passenger flows of any two stations. Five types of spatial blocks are designed (Fig. 3). Each small box represents a single graph convolution for the three graph elements W_A , W_S and W_C , as described in equation (2). The passenger flow of all stations in a metro network X_t is the input of the spatial block, and the output (H_S) is the hidden state in the last layer of the GCN.

The weight matrix and the construction mode of multi-graph convolution that are selected for each of the five types of spatial blocks defined in Fig. 3 are different. Fig. 3(a), Fig. 3(b), and Fig. 3(c) represent the construction forms of a single simple graph convolution that only considers the station adjacency, station passenger flow pattern similarity, and passenger flow relationship between stations, respectively. Fig. 3(d) is a multi-graph convolutional pattern that connects three graphs (station adjacency graph, passenger flow similarity graph, and passenger flow correlation graph) by a parallel structure. Fig. 3(e) is a multi-graph convolutional pattern similar to that in Fig. 3(d) but with the three graphs stacked rather than parallel. In Fig. 3, $\oplus$ denotes a feature concatenation operation, LN denotes normalization, and ϕ denotes the nonlinear activation function (ReLU).

4.3. Building time blocks

As an alternative to LSTM and RNN, the GRU can nonlinearly manage the dynamics of historical data for metro passenger flow and learn intricate time-dependent relationships (Fu et al., 2016). Additionally, GRU learns relatively long-term dependencies (Cho et al., 2014) with the reset gate R_t , the input gate Z_t , the updated data N_t , and the new output H_t (the hidden state in the last layer of the GRU). Therefore, the GRU is built to learn the time dependency of metro passenger flow, and its structure is shown in Fig. 4.

Specifically, the reset gate R_t , the input gate Z_t , the updated data N_t , and the new output H_t can be calculated using equation (9).

$$\begin{cases} R_t = \text{sigmoid}(W_{rx} * X_t + W_{rh} * H_{t-1} + b_r) \\ Z_t = \text{sigmoid}(W_{zx} * X_t + W_{zh} * H_{t-1} + b_z) \\ N_t = \tanh(W_{nx} * X_t + R_t \odot (W_{nh} * H_{t-1} + b_h)) \\ H_t = (1 - Z_t) \odot N_t + Z_t \odot H_{t-1} \end{cases} \quad (9)$$

In the GRU, the hidden state of the last iteration $t-1$ (H_{t-1}) and the passenger flow data (X_t) are used as inputs. *Sigmoid* and *tanh* are activation functions. W_{rx} denotes the parameters between R_t and passenger

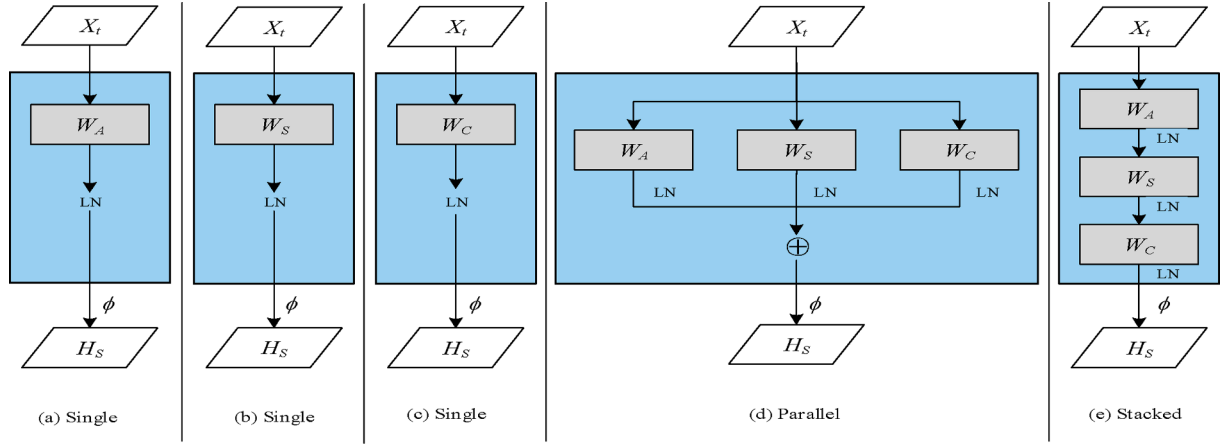


Fig. 3. Five types of spatial blocks.

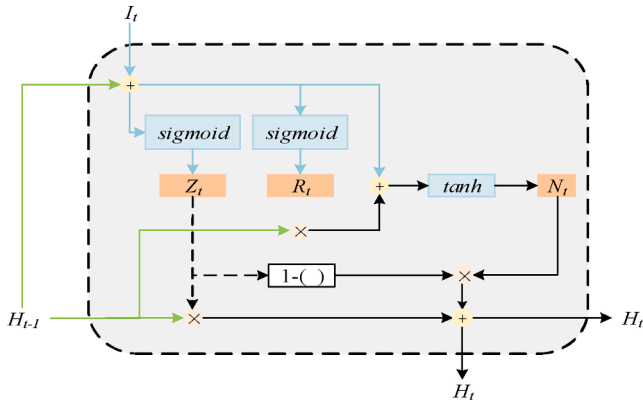


Fig. 4. GRU structure.

flow data X_t , and W_{rh} represents the parameters between R_t and H_{t-1} . Other parameters W_{zx} , W_{zh} , W_{nx} , and W_{nh} share a similar function that needs to be learned; b_r , b_z , and b_h represent bias terms. The temporal characteristics from the metro passenger flow data can be effectively captured using GRU.

4.4. ASC-GRU based spatial-temporal prediction

Most studies follow a straightforward sequence of feeding the output of the spatial blocks into the temporal blocks to facilitate traffic prediction (Cui et al., 2019; Jin et al., 2020). However, using such a sequential structure for feature extraction may result in feature distortions during the conversion of the convolution results, potentially leading to information loss and uncertain predictions (Ma et al., 2018). To address this limitation, in this study, the results of the spatial and temporal blocks are integrated to complement each other, and a parallel combination is built to achieve a more accurate prediction of metro passenger flow. This approach aims to leverage the strengths of both spatial and temporal modules to provide a more comprehensive and robust prediction model for metro passenger flow.

As shown in Fig. 1, first, the station feature values (passenger flows X_t) are input into the spatial and temporal blocks to learn the spatial and temporal features. The output results of the spatial and temporal blocks are H_S and H_T , respectively. Then, H_S and H_T are used to generate a comprehensive hidden state $\hat{H}$ with a fully connected layer (FC) using equation (10), where $\oplus$ denotes a feature concatenation operation.

$$\hat{H} = FC(H_S \oplus H_T) \quad (10)$$

The final prediction is obtained using the fully connected layer as in

equation (11), where w and b denote the trainable weight and the bias, respectively.

$$\hat{X} = w\hat{H} + b \quad (11)$$

5. Experiments

This section describes the datasets used in the experiment. It presents the validation of the spatial-temporal dependency of metro passenger flow using real-world data, specific implementation details, comparison with other baseline models, and analysis of the usefulness of the components introduced into the proposed model.

5.1. Data description

5.1.1. Data source

Passenger travel records, also known as passenger flow data were collected from the Hangzhou and Shanghai metro systems in China. Experiments were conducted on these two large-scale datasets to verify the effectiveness of the proposed approach. For the Hangzhou metro dataset, data were collected from January 1, 2019, to January 25, 2019, with a daily passenger flow reaching 2.35 million riders. The Hangzhou metro network consists of 80 stations and 248 connecting edges. The experimental data for the Hangzhou metro is divided into three parts: the first part, from January 1 to January 18 (18 days), is used for training; the data from January 19 to January 20 (2 days) are used as a validation set to select the best model during the training process; and the remaining data over 5 days, from January 21 to January 25, are used as a test set to evaluate the generalization ability of the final model. The Shanghai metro network's data were collected from July 1, 2016, to September 30, 2016, with a daily passenger flow of 8.82 million riders. The Shanghai metro network consists of 288 stations and 958 edges in the test case. The experimental data for the Shanghai metro also comprises three segments: the training set covers the first 62 days (July 1 to August 31), the subsequent 9 days (September 1 to September 9) serve as the validation set; and the test set spans 21 days (September 10 to September 30). For more detailed information regarding the original data utilized in this study, please refer to Liu et al., (2020).

The raw transaction record of a passenger trip includes the smartcard ID, the station ID, the station name, the line ID, the card swiping time, and the entry or exit gate ID. This raw transaction record originates from an automatic fare collection (AFC) system. The preprocessed data are used for flow prediction. We specifically select the daily operation period from 06:00 to 23:30. During the data preprocessing step, we filter out the records collected beyond the operation period. We then remove unreasonable records, such as outbound time being less than inbound time or traveling across days. To fill in the missing values, linear

interpolation is used. The data is aggregated every 15 min, resulting in 70 samples per day, and the passenger flow for each station sample is calculated.

The information on the two datasets is shown in Table 1, where “HZMetro” and “SHMetro” represent the metro datasets of Hangzhou and Shanghai, respectively.

5.1.2. Analysis of data spatial-temporal characteristics

As mentioned earlier, in addition to the local spatial dependency of neighboring stations, passenger flow has other spatial characteristics, such as passenger flow similarity and passenger flow correlation. In this subsection, the HZMetro dataset is analyzed to validate the presence of these spatial characteristics in real-world data. Furthermore, the appropriate time intervals used in metro passenger flow prediction are analyzed according to the HZMetro dataset. The correlation coefficient (Pearson coefficient) is used to measure the connection degree between the passenger flow patterns of two stations, with a correlation coefficient close to 1 indicating a strong similarity in passenger flow patterns between the two stations.

An internet search reveals 20 primary POI categories for Hangzhou city, including shopping centers, real-estate properties, and tourist attractions. As passenger flow is influenced by travel attraction, the function of the area in which the metro station is located affects the spatial characteristics of passenger flow, leading to various spatial distribution patterns, including non-peak, full-peak, and double-peak passenger flow distributions. Three sample stations (Liangzhu, Longxiang Qiao, and Hangzhou Hair Factory) located in areas with different functions are selected for this analysis, representing various passenger flow patterns within the Hangzhou metro network (Fig. 5). Liangzhu station, located in an area with natural features, serves a region with low development and low passenger flow, resulting in a no-peak passenger flow pattern. Longxiang Qiao station, situated in a tourist function area, serves a highly developed region with a large passenger volume throughout the day, leading to a full-peak passenger flow pattern. Hangzhou Hair Factory, located in a real-estate functional area, serves a region with a large volume of passenger inflow during the morning and outflow during the evening rush hour, resulting in a double-peak passenger flow pattern. Analysis of these three example stations reveals that stations serving areas with different functions tend to exhibit different spatial characteristics in passenger flow patterns.

As shown in Fig. 5, the Hangzhou Hair Factory and Feihong Road stations are not directly adjacent to each other on the metro network, but their locations share the same real-estate role. Analysis of the inbound and outbound passenger flows at Hangzhou Hair Factory and Feihong Road reveals that the stations exhibit similar passenger outflow/inflow patterns, specifically double-peak patterns, on Jan. 7, 2019 (Monday) (Fig. 6). Additionally, the correlation coefficient of passenger flow between the Hangzhou Hair Factory and Feihong Road stations is ~ 0.83 ($\sim$ means approximately), indicating an extremely strong correlation between the passenger flow patterns of the two stations. This finding demonstrates that the passenger flow of two non-adjacent stations exhibits a spatial similarity feature. Therefore, constructing a passenger flow similarity graph is necessary to account for

these spatial similarities and improve the accuracy of metro passenger flow prediction.

The West Lake Culture Square station and the Jiangling Road station on Hangzhou Metro Line 2 (as seen in Fig. 5) are not adjacent to each other. However, the passenger flow records between both stations account for a significant portion ($\sim 5\%$) of the total passenger flow records. These two stations display similar passenger outflow/inflow patterns, specifically full-peak patterns (Fig. 7). The correlation coefficient of passenger flow between the two stations is ~ 0.86 , indicating an extremely strong correlation between their passenger flow patterns. Even when two stations are not adjacent on a metro line, their passenger flows exhibit a strong spatial feature correlation when a significant passenger flow exists between them. This demonstrates the importance of building a passenger flow correlation graph for metro passenger flow prediction.

We have selected the preprocessed inflow data for the Wulin Square station of the Hangzhou metro network from January 7, 2019, to January 27, 2019, three consecutive weeks from Monday to Sunday. A visualization of the inflow data is shown in Fig. 8. Based on the below figure, we note that the inflow has a certain periodicity. The distributions of the inbound passenger flow on weekdays (Monday to Friday) and weekends (Saturday and Sunday) are inconsistent. As described in Fig. 9, a peak of passenger flow occurs on weekdays. The distributions of the inflow on weekdays (Monday to Friday) is strongly correlated for three consecutive weeks. And the distribution of the inflow on weekends is highly similar for the three consecutive weeks as described in Fig. 10.

Additionally, the correlation coefficients between the current passenger flow at time step t and the passenger flows at the previous eight time steps (each step representing 15 min) are calculated, and the results are presented in Table 2. The correlation coefficient increases as the time interval gets closer to the current period (t), indicating a temporary correlation in passenger flow that varies with different time intervals. As a rule of thumb, if the correlation coefficient of two variables exceeds 0.8, these two variables are considered highly correlated; otherwise, they are not highly correlated. For the periods from $t-5$ to $t-8$, the correlation coefficient is lower than 0.8, indicating that passenger flows in these periods have a smaller influence on the passenger flow of the current period compared with other periods. Consequently, choosing reasonable prediction time intervals (e.g., 15, 30, 45, and 60 min) can improve prediction accuracy, while time intervals longer than 60 min may not provide significant benefits. These intervals of 15, 30, 45, and 60 min are selected for predicting metro passenger flow in this study.

5.2. Experimental settings

5.2.1. Evaluation metrics

To evaluate the model performance, the commonly used error metrics are adopted, namely root-mean-square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE), and they are defined by equations (12)–(14), respectively.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{X}_i - X_i)^2} \quad (12)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{X}_i - X_i| \quad (13)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|\hat{X}_i - X_i|}{X_i} \quad (14)$$

Here, n denotes the number of samples in experiments, X_i is the actual value, and $\hat{X}_i$ is the predicted value (normalized). A smaller value of the three indicators (i.e., RMSE, MAE, and MAPE) corresponds to better model prediction performance. According to the analysis in Section 5.1, the prediction time interval is divided into 15, 30, 45, and 60 min, and the error values of the indicators are calculated for each of the four time-interval cases.

Table 1
Details of the datasets.

Dataset	HZMetro	SHMetro
Time span	2019/1/1–2019/1/25	2016/7/1–2016/9/30
Time interval	15 min	15 min
Samples per day	70	70
Station	80	288
Physical edge	248	958
Ridership per day	2.35 million	8.82 million
Train timespan	2019/1/1–2019/1/18	2016/7/1–2016/8/31
Validation timespan	2019/1/19–2019/1/20	2016/9/1–2016/9/9
Test timespan	2019/1/21–2019/1/25	2016/9/10–2016/9/30



Fig. 5. Distribution of stations in the Hangzhou metro system and three passenger flow patterns.

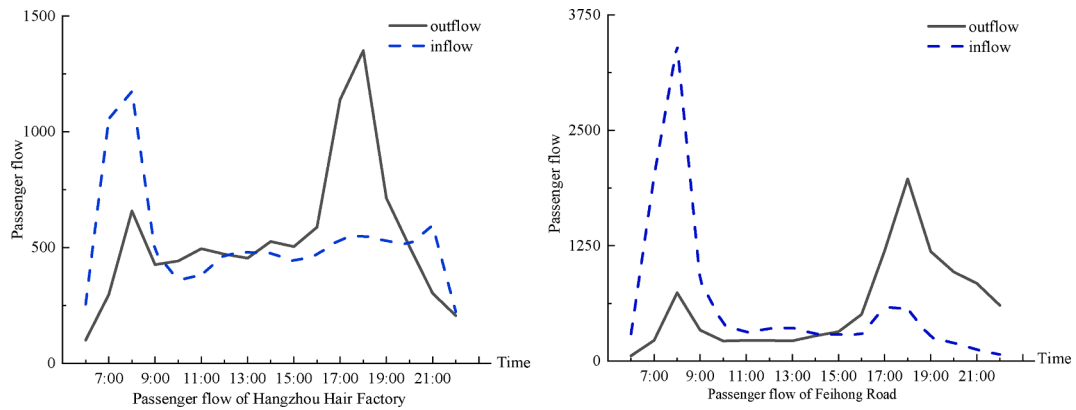


Fig. 6. Passenger patterns of the Hangzhou Hair Factory station and the Feihong Road station.

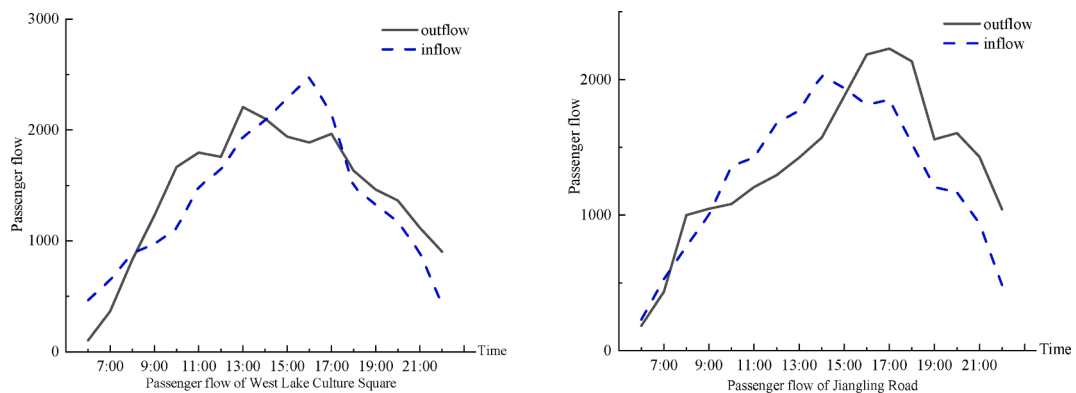


Fig. 7. Passenger patterns of the West Lake Culture Square station and the Jiangling Road station.

5.2.2. Experimental parameters

The SHMetro dataset contains a large number of stations, 288, and thus the computational costs are high. To simplify the calculation, Liu et al. (2020) selected only 10 stations with high similarity as the primary

research objects when defining the passenger flow similarity graph. Similarly, they selected the top 10 stations with high correlation rates to construct the passenger flow correlation graph. The same station selection principle as in Liu et al. (2020) is adopted in this study to

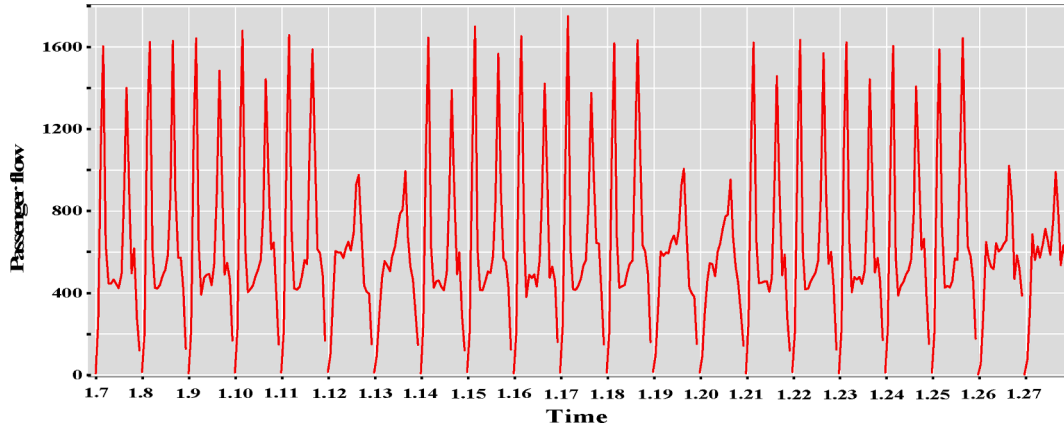


Fig. 8. Inflow of the Wulin Square station.

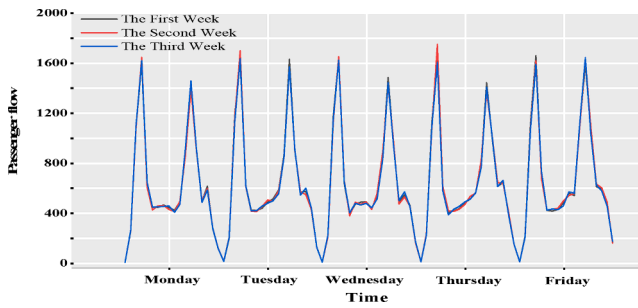


Fig. 9. Inflow on weekdays of the Wulin Square station.

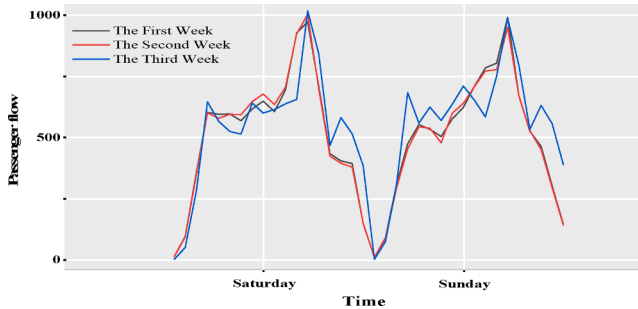


Fig. 10. Inflow on weekends of the Wulin Square station.

Table 2

Pearson coefficients of passenger flow at different time intervals.

Time interval	Pearson coefficient	Time interval	Pearson coefficient
$t-1$ (0–15 min)	0.96	$t-5$ (60–75 min)	0.79
$t-2$ (15–30 min)	0.92	$t-6$ (75–90 min)	0.75
$t-3$ (30–45 min)	0.88	$t-7$ (90–105 min)	0.73
$t-4$ (45–60 min)	0.86	$t-8$ (105–120 min)	0.69

construct the passenger flow similarity graph and the passenger flow correlation graph. Although there are 288 stations in total, only 2880 (288×10) edges are considered in constructing the passenger flow similarity and correlation graphs. The HZMetro dataset comprises 80 stations, which is much smaller than that of the SHMetro dataset. Thus, the estimation process involving the HZMetro is less computationally demanding than that involving the SHMetro dataset.

The proposed model was implemented using the PyTorch framework, and the model comparison experiments were conducted in an environment with GeForce RTX 3090 running Ubuntu 18.04. All of the

baseline models were implemented or imported using Python from existing packages. Before input, the data in both datasets were normalized via Z-score normalization to enhance the model's accuracy. To optimize the model's performance, parameter adjustment was conducted. The input and output sequence lengths were set to 4, the feature dimension was set to 256, the learning rate was set to 0.001, the decay rate was set to 0.1, and the number of networks was set to 200 using the Adam optimization algorithm. For each benchmark, the proposed ASC-GRU model was trained using the training set, and the model's hyperparameters were determined using the validation set.

5.2.3. Benchmark methods

The proposed model is compared with several other benchmark models, most of which are traditional neural network models and deep learning network models. The details of the baseline models are given as follows:

- Historical average (HA): It uses the average ridership of the corresponding historical periods to predict passenger flow.
- Random forest (RF): This machine learning approach constructs numerous decision trees to tackle regression and classification problems.
- Gradient boosting decision trees (GBDT): It is a weighted ensemble algorithm that mainly uses additive models and forward stepwise algorithms to achieve learning.
- Multiple layer perceptron (MLP) (Chen et al., 2021): It is the most basic deep learning method and consists of input layers, hidden layers, and output layers.
- Long short-term memory (LSTM) (Liu et al., 2020): It contains input, forget, and output gates for time-series forecasting problems.
- Attention-based spatial-temporal graph convolutional networks (ASTGCN) (Guo et al., 2019): It incorporates spatial-temporal graph convolution and attention mechanisms to capture spatial and temporal characteristics of traffic data for traffic forecasting.
- Graph-WaveNet (Wu et al., 2019): It uses superimposed and extended one-dimensional convolution to capture temporal correlations and establishes an adaptive matrix to capture the spatial dependency. Then, temporal modeling and spatial modeling are fused for passenger flow forecasting.
- AGCRN (Bai et al., 2020): It does not require a predefined spatial graph and is an adaptive graph convolutional network that can learn spatiotemporal dependencies from passenger flow data.
- PVCN (Liu et al., 2020): It constructs three topological graphs (the physical graph, the similarity graph, and the correlation graph) to capture spatial correlations in the metro system, and proposes a physical-virtual collaboration graph network (PVCN) to predict passenger flow.

- **STDGRL** (Xie et al., 2022): It employs a dynamic graph relationship learning module to learn dynamic spatial relationships between metro stations without a predefined graph adjacency matrix, and provides a transformer-based long-term relationship prediction module for passenger flow prediction.

5.3. Results and analysis

The performances of the proposed model and other benchmark models are compared using the HZMetro and SHMetro datasets, and the results are shown in Tables 3 and 4. Our proposed model ASC-GRU (stacked) achieves the best performance in most cases. Specifically, the proposed model achieves average improvements of 4.85 % and 7.09 % in MAPE on the Hangzhou and Shanghai Metro datasets compared with the baseline models at different time intervals.

HA is a mathematical-statistical model. RF and GBDT are traditional machine learning models. These three models have significant drawbacks. Among the benchmark models, HA exhibits the largest MAPE values in the 15 min time interval for both the HZMetro and SHMetro datasets: 19.14 % and 31.55 %, respectively. Moreover, the HA model features large prediction errors for the other three time-interval cases (30, 45, and 60 min), with poor but relatively stable predictions (stable at approximately 20 % and 30 % on average for the HZMetro and SHMetro datasets, respectively). For the 15 min time-interval case, the RF and GBDT models exhibit smaller MAPE values (17 %–18 % on the HZMetro dataset and 23 %–24 % on the SHMetro dataset) than HA. However, as the time interval increases, the MAPE values for the RF and GBDT models sharply increase and become larger than those of HA. For the HZMetro dataset, when the time interval increases to 45 min, the average value reaches ~ 26 %; when the time interval increases to 60 min, the average value reaches ~ 38 %. The prediction performance continuously deteriorates because the RF and GBDT models only explore the time-dependent characteristics of passenger flow. Moreover, RF and GBDT show limited learning efficiency on the time series of passenger flow, leading to degraded prediction quality as the length of the time series increases. For the HZMetro dataset, the average MAPE for HA, RF, and GBDT reaches ~ 31.74 % over the 60 min interval, and the proposed model (ASC-GRU) achieves an average MAPE reduction of 14.61 % over the 60 min interval compared with the above three traditional models. Compared with the first three models (HA, RF, and GBDT), the ASC-GRU model achieves average improvements of 36.07 in RMSE and 20.75 in MAE over the 60 min interval. Therefore, the proposed model exhibits significantly improved performance in terms of the MAPE, RMSE, and MAE values.

Compared with the first three conventional models (HA, RF, and GBDT), LSTM and MLP exhibit significantly smaller values for all of the error metrics. This indicates that LSTM and MLP have a substantially improved ability to capture time features compared with the three

conventional methods. Both models possess particular automatic learning abilities for the time series data; however, this ability does not have a significant self-improvement effect on error values at different time intervals. Therefore, their prediction accuracy decreases as the time interval increases, resulting in lower stability for both two models. In addition, the LSTM and MLP models only capture the temporal dependencies but do not consider spatial features. Simultaneously considering temporal and spatial features enables the extraction of comprehensive spatial-temporal characteristics, thereby significantly improving prediction accuracy. The proposed ASC-GRU model features significantly improved prediction performance compared with the LSTM and MLP models because it considers both temporal and spatial features.

For the HZMetro dataset with a 60 min interval, ASC-GRU achieves average improvements of 4.09 % in MAPE, 4.08 in RMSE, and 2.38 in MAE compared with the LSTM and MLP models. For the SHMetro dataset with a 15 min interval, the ASC-GRU model achieves a 1.18 % lower MAPE on average than the LSTM and MLP models. Furthermore, the ASC-GRU achieves a 4.52 % lower MAPE on average than the LSTM and MLP models at a 60 min interval. The results show that ASC-GRU achieves greater performance improvement than the LSTM and MLP models as the time interval increases. Although the LSTM and MLP models can capture the temporal characteristics of passenger flow, ASC-GRU considers both temporal and spatial characteristics, allowing it to improve the performance of long-term prediction.

Although the MLP model yields better (i.e. lower) MAE and RMSE values for the 30 min interval case than the proposed ASC-GRU model on the SHMetro dataset, the ASC-GRU model remains a valid approach. The MLP model has some advantages at short time intervals. Its MAE and RMSE are only 0.14 and 0.32 less than those of the proposed ASC-GRU model, respectively, and thus, its advantage over the proposed model is not significant. However, as the time interval increases, the ASC-GRU model shows a significant advantage in terms of MAE and RMSE (Fig. 11). The MAPE value is better than the MAE and RMSE values because MAPE can be more easily interpreted and is less likely to be affected by extreme values. The MLP model has little advantage over the ASC-GRU model in terms of MAPE (Fig. 12(b)). For the 60 min interval, the ASC-GRU model achieves a 3.20 % improvement in MAPE compared with MLP for the SHMetro dataset.

The ASTGCN and Graph-WaveNet models can capture the spatial characteristics of passenger flow. However, the error values of these two models increase significantly as the time interval increases (Fig. 12(a) and Fig. 12(b)). Both the ASTGCN and Graph-WaveNet models exhibit worse performance on the SHMetro dataset than the MLP model. In particular, at a prediction time interval of 60 min, ASTGCN exhibits a ~ 10 % higher MAPE value than the MLP model. Furthermore, with the increase in the time interval from 15 to 60 min, the MAPE values of ASTGCN and Graph-WaveNet models on the SHMetro dataset increase by 10.42 % and 4.69 %, respectively, while that of the ASC-GRU model

Table 3
Model performance comparison using the HZMetro dataset.

Model	15 min			30 min			45 min			60 min			Average		
	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE
HA	19.14	36.37	64.19	19.31	36.37	64.10	19.57	36.23	63.92	20.01	35.99	63.72	19.51	36.24	63.98
RF	18.34	32.19	53.52	21.46	38.00	54.00	26.51	45.78	80.06	37.12	53.95	94.29	25.86	42.48	70.47
GBDT	17.60	30.88	51.50	20.49	36.48	61.94	25.75	44.12	76.70	38.10	51.10	91.21	25.49	40.65	70.34
LSTM	14.41	25.76	45.30	15.54	26.01	45.52	17.18	26.38	46.3	19.88	26.76	47.53	16.75	26.23	46.16
MLP	16.26	26.57	46.55	17.10	27.44	47.96	19.01	28.79	50.66	22.56	30.52	54.62	18.73	28.33	49.95
ASTGCN	15.05	27.34	46.19	15.56	27.74	46.16	16.48	28.20	46.70	17.75	28.85	49.70	16.21	28.03	47.19
Graph-WaveNet	14.27	24.07	41.88	15.23	25.48	43.70	17.34	27.15	46.50	19.37	29.14	48.69	16.55	26.46	45.19
AGCRN	23.35	23.62	40.35	25.44	24.94	43.19	26.47	25.95	45.28	31.34	27.40	46.78	26.65	25.48	43.90
PVCGN	14.03	24.48	40.76	15.27	25.33	43.37	16.46	26.39	45.48	17.55	27.28	48.35	15.83	25.87	44.49
STDGRL	20.91	23.27	39.55	21.41	23.77	40.43	22.30	24.89	42.88	25.70	25.83	45.18	22.58	24.44*	42.01*
ASC-GRU	14.22	24.23	41.36	14.95	25.36	43.26	15.97	25.91	46.07	17.13	26.26	47.00	15.57*	25.44	44.42

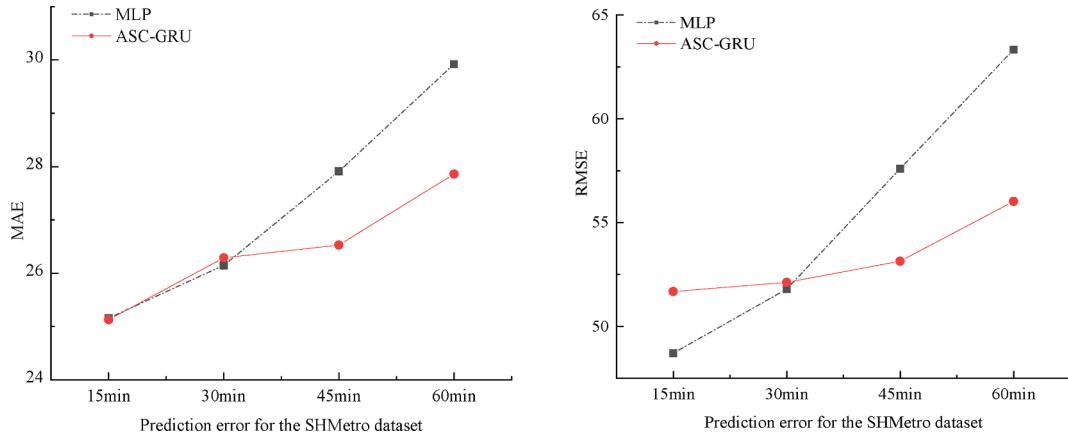
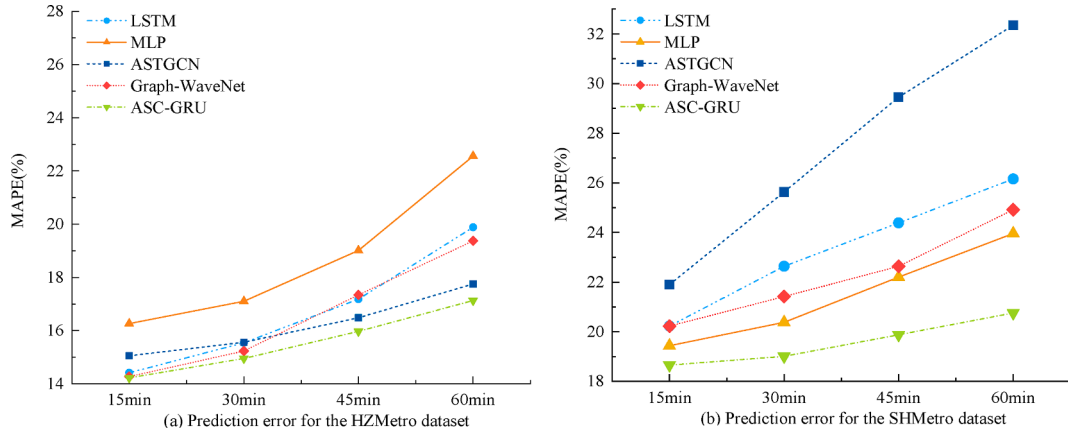
Note: * represents the best result.

Table 4

Model performance comparison using the SHMetro dataset.

Model	15 min			30 min			45 min			60 min			Average		
	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE
HA	31.55	48.26	136.97	31.49	47.88	136.81	31.27	47.26	136.45	30.80	46.40	135.72	31.28	47.45	31.55
RF	24.09	34.37	66.63	28.89	41.37	88.03	41.34	50.91	118.65	52.91	59.15	143.50	36.81	46.45	24.09
GBDT	23.40	32.72	62.59	28.17	39.50	82.32	40.76	49.14	113.95	52.60	57.31	137.50	36.23	44.67	23.40
LSTM	20.23	26.68	55.53	22.64	27.25	57.37	24.39	28.08	60.45	26.16	28.94	63.41	23.36	27.74	20.23
MLP	19.44	25.16	48.71	20.38	26.15	51.80	22.20	27.91	57.60	23.96	29.92	63.33	21.50	27.29	19.44
ASTGCN	21.90	32.29	66.49	25.63	39.28	98.76	29.45	46.59	133.28	32.35	51.37	154.95	27.33	42.38	21.90
Graph-WaveNet	20.23	24.91	47.73	21.42	27.12	54.15	22.64	29.23	60.10	24.92	31.56	68.10	22.30	28.21	20.23
AGCRN	23.16	24.01	47.11	24.70	25.46	50.96	26.47	27.04	55.57	26.96	28.41	59.61	25.32	26.23	23.16
PVCGN	18.93	24.55	52.82	19.76	26.14	53.43	20.69	28.42	55.59	21.74	29.39	57.64	20.28	27.13	18.93
STDGRL	21.43	23.73	46.87	21.66	24.38	49.29	22.48	25.42	52.90	23.41	26.58	57.40	22.25	25.03*	21.43
ASC-GRU	18.66	25.13	51.68	19.01	26.29	52.12	19.88	26.53	53.14	20.76	27.86	56.02	19.58*	26.45	18.66*

Note: * represents the best result.

**Fig. 11.** MAE and RMSE curves of MLP and ASC-GRU models on the SHMetro dataset.**Fig. 12.** MAPE curves of five models on the two metro datasets.

increases by less than 3 %. Therefore, the ASC-GRU model achieves a more stable prediction performance, with average improvements of 1.12 % in MAPE, 4.56 in RMSE, and 1.94 in MAE compared with the ASTGCN and Graph-WaveNet models over the 60 min interval. Although both the ASTGCN and Graph-WaveNet models incorporate graph convolution, they only use physical topology graphs as graph elements to represent the direct connections between two metro stations. In contrast, the ASC-GRU model constructs passenger flow similarity and correlation graphs, thereby achieving a more advanced performance by adequately characterizing the complex passenger flow patterns within

the metro system compared with the ASTGCN and Graph-WaveNet models.

The Graph-WaveNet model exhibits better MAE and RMSE values at 15 min intervals than the proposed ASC-GRU model on the SHMetro dataset. At short time intervals, Graph-WaveNet features some advantages in terms of MAE and RMSE (Fig. 13). However, the ASC-GRU model shows a significant advantage at longer time intervals. Additionally, the ASC-GRU model achieves more convincing MAPE values (Fig. 12(b)). At the 60 min interval, the ASC-GRU model demonstrates ~ 12 % improvement in MAPE compared with the Graph-WaveNet

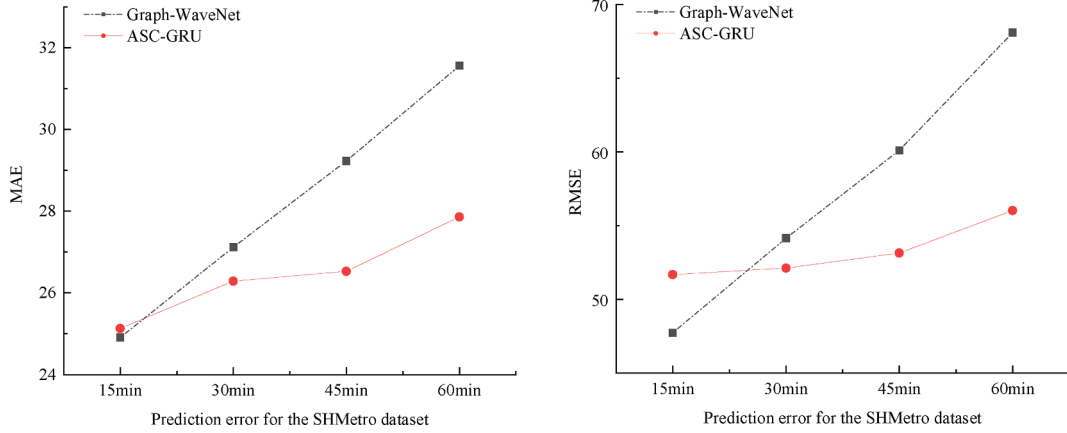


Fig. 13. MAE and RMSE curves of Graph-WaveNet and ASC-GRU on the SHMetro dataset.

model on the SHMetro dataset.

There is little difference in performance between the PVCGRN and ASC-GRU model for the HZMetro dataset. However, the ASC-GRU model outperforms the PVCGRN model in terms of MAPE, MAE, and RMSE for the SHMetro dataset. In the case of the SHMetro dataset, ASC-GRU achieves average improvements of 0.70 % in MAPE, 0.67 in MAE, and 1.63 in RMSE compared to the PVCGRN model. The results indicate that combining the spatial module and the temporal module in a parallel manner enhances the accuracy of metro passenger flow prediction as opposed to directly embedding the spatial module into the temporal module.

The MAE and RMSE values of the AGCRN and ASC-GRU models are identical. However, our model exhibits a significant advantage in MAPE compared to the AGCRN model. For the HZMetro dataset, the ASC-GRU model achieves an average improvement 11.08 % in MAPE compared to the AGCRN model, and for the SHMetro dataset, it achieves an average improvement 5.75 % in MAPE compared to the AGCRN model. While the STDGRL model has an advantage over our proposed ASC-GRU model in terms of MAE and RMSE, its advantage is not significant. The STDGRL model achieves an average improvements of 2.41 in RMSE and 1.00 in MAE for the HZMetro dataset and an average improvements of 2.63 in RMSE and 1.43 in MAE for the SHMetro dataset compared to the ASC-GRU model. However, our ASC-GRU model consistently outperforms the STDGRL model in terms of MAPE. For the HZMetro dataset, the ASC-GRU model achieves a 7.01 % average improvement in MAPE compared to the STDGRL model, and for the SHMetro dataset, it achieves a 2.67 % average improvement in MAPE compared to the STDGRL model. MAPE is a more convincing metric than the MAE and RMSE since it is easier to interpret and is less likely to be affected by extreme values. Therefore, these comparisons demonstrate that our proposed ASC-GRU model can achieve higher prediction accuracy.

Generally, all of the models exhibit deteriorating prediction performance with increasing time intervals, indicating that prediction regularities can be more easily captured at shorter time intervals. However, among the models, the ASC-GRU model exhibits notable performance at longer time intervals and larger datasets, such as the SHMetro dataset. Specifically, the model shows an average improvement of 1.79 % compared with the other four models (LSTM, MLP, ASTGCN, and Graph-WaveNet) on the SHMetro dataset with a 15 min interval, and this improvement increases to 6.09 % at a longer time interval of 60 min. Thus, the ASC-GRU model outperforms other baseline models in predicting over long time intervals.

5.4. Performance comparison of variant models

In the previous section, it is verified that the proposed ASC-GRU model outperforms other baseline models in passenger flow

prediction. In this section, the prediction performances of several variant models constructed via the multi-graph convolution method are compared to verify the usefulness of each proposed component. Several model variants are introduced as follows:

- **A-GRU**: This variant uses only the station adjacency graph and the GRU to predict metro passenger flow.
- **S-GRU**: This variant uses only the passenger flow similarity graph and the GRU to predict metro passenger flow.
- **C-GRU**: This variant uses only the passenger flow correlation graph and the GRU to predict metro passenger flow.
- **ASC-GRU (parallel)**: In this variant, the station adjacency graph, passenger flow similarity graph, and passenger flow correlation graph are constructed in parallel and then combined with the GRU in parallel for metro passenger flow prediction.
- **ASC-GRU (stacked)**: The complete model proposed in this study (i. e., ASC-GRU), in which the station adjacency graph, passenger flow similarity graph, and passenger flow correlation graph are constructed in a stacked manner and then combined with the GRU in parallel to predict metro passenger flow.

Experiments were conducted on the HZMetro and SHMetro datasets to compare the prediction performances of the mentioned variant models. The results are shown in Table 5 and Table 6 and Fig. 14 and Fig. 15. ASC-GRU (parallel) and ASC-GRU (stacked) simplify to ASC-GRU-P and ASC-GRU-S in the figures and tables below. The proposed model, ASC-GRU-S (ASC-GRU), outperforms the other variant models across all time-interval cases on both the HZMetro and SHMetro datasets.

For the four time intervals, the S-GRU model features the largest MAPE value among the three single models, and its performance worsens with increasing time intervals. At a short time interval (15 min), no significant difference exists between the MAPE values of the three single models: 14.85 % for S-GRU, 14.82 % for A-GRU, and 14.72 % for C-GRU. As the time interval increases to 30 and 45 min, the difference in error values between S-GRU and the other two models increases by ~ 1 % on average. However, as the time interval increases to 60 min, the difference in error values reaches ~ 2 % on average. The results obtained using the SHMetro dataset show a similar trend. Therefore, as the time interval becomes longer, the performance of the S-GRU model degrades. Among the three single models, the C-GRU model exhibits the best performance (i.e., with the lowest MAPE value), indicating the usefulness of the passenger flow correlation graph. Although distance is a commonly used base graph element, the passenger flow correlation graph is the most useful for representing complex spatial dependencies.

The ASC-GRU (parallel) and ASC-GRU (stacked) models outperform the three single graph models (i.e., A-GRU, S-GRU, and C-GRU). The

Table 5

Performance comparison of variant models for the HZMetro dataset.

Model	15 min			30 min			45 min			60 min		
	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE
A-GRU	14.82	25.86	42.70	15.90	26.33	44.99	17.09	28.30	49.30	18.84	30.67	56.64
S-GRU	14.85	26.53	43.15	16.29	27.39	46.32	17.66	29.33	50.28	20.23	31.79	56.36
C-GRU	14.72	25.37	42.16	15.76	25.85	44.16	16.82	27.34	47.90	18.24	29.72	50.77
ASC-GRU-P	14.63	24.93	41.92	15.44	25.36	43.26	16.51	27.00	47.00	18.04	29.00	50.00
ASC-GRU-S	14.22*	24.23*	41.36*	14.95*	25.00*	43.00*	15.97*	25.91*	46.07*	17.13*	26.26*	47.00*

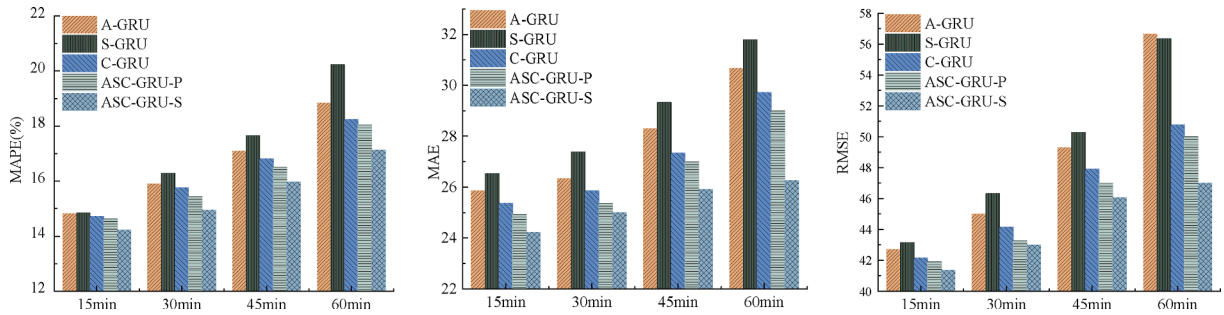
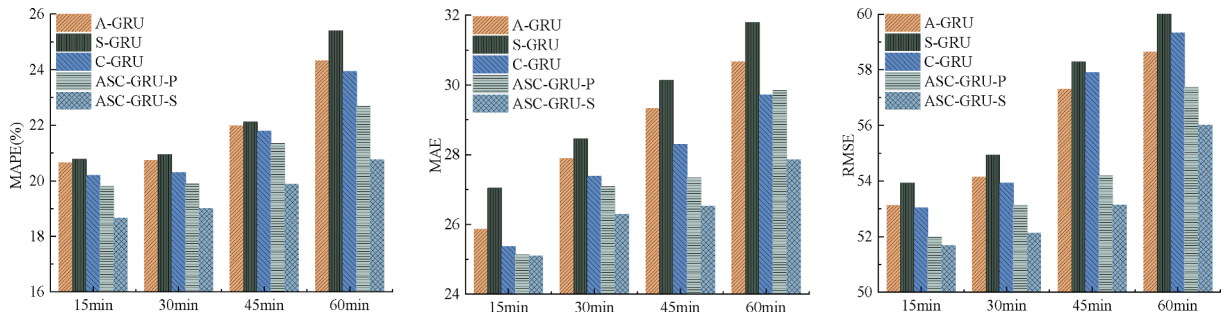
Note: * represents the best result.

Table 6

Performance comparison of variant models for the SHMetro dataset.

Model	15 min			30 min			45 min			60 min		
	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE
A-GRU	20.65	25.86	53.12	20.74	27.89	54.14	21.98	29.33	57.30	24.32	30.67	58.64
S-GRU	20.77	27.04	53.92	20.95	28.46	54.93	22.12	30.13	58.28	25.41	31.79	60.36
C-GRU	20.19	25.37	53.04	20.30	27.39	53.92	21.79	28.30	57.90	23.93	29.72	59.33
ASC-GRU-P	19.80	25.13	51.99	19.89	27.09	53.12	21.35	27.34	54.19	22.69	29.85	57.37
ASC-GRU-S	18.66*	25.10*	51.68*	19.01*	26.29*	52.12*	19.88*	26.53*	53.14*	20.76*	27.86*	56.02*

Note: * represents the best result.

**Fig. 14.** Performance comparison of five variant models for the HZMetro dataset.**Fig. 15.** Performance comparison of five variant models for the SHMetro dataset.

ASC-GRU (parallel) model achieves average improvements of 0.81 % in MAPE, 2.21 in RMSE, and 1.22 in MAE compared with the three single graph models at different time intervals. Similarly, the ASC-GRU (stacked) model achieves average improvements of 1.78 % in MAPE, 3.27 in RMSE, and 2.29 in MAE compared with the three single graph models at different time intervals. These experimental results illustrate the complementary nature of the graphs, and the newly added elements significantly contribute to the better description of the spatial feature relationships.

Moreover, the ASC-GRU (stacked) model outperforms the ASC-GRU (parallel) in terms of prediction accuracy. The former achieves average improvements of 0.97 % in MAPE, 1.06 in RMSE, and 1.07 in MAE

compared with the latter at different time intervals, suggesting that the stacked structure demonstrates higher effectiveness than the parallel structure in metro passenger flow prediction.

6. Conclusion

This study introduces an ASC-GRU framework to obtain a more accurate metro passenger flow (inflow and outflow at each station of a metro network) prediction. First, three spatial dependencies (station adjacency, passenger similarity, and passenger correlation) between stations in the metro network are selected. Then, stacked spatial blocks with spatial graph elements are built. Finally, a parallel integration of

spatial blocks and temporal components composed of GRU is used to create a joint prediction model, capturing the spatial-temporal characteristics of metro passenger flow.

To evaluate the accuracy and generality of the proposed method, experiments are conducted using real datasets from the Hangzhou and Shanghai metro systems in China. Various types of baseline models are selected for performance comparison. The experimental results demonstrate that the proposed method achieves a more accurate passenger flow forecast than other baseline models. Specifically, the proposed model outperforms the baseline models at different time intervals, with 4.30 % and 8.82 % lower MAPE on average on the Hangzhou and Shanghai Metro datasets, respectively. Additionally, comparison experiments on the prediction effectiveness of several model variants reveal that of all of the variants, ASC-GRU (stacked) (the proposed model) yields the best results, with an MAPE 1.78 % less than that of the three single models on average at different time intervals. These results demonstrate the effectiveness of the proposed model in achieving accurate and reliable metro passenger flow prediction. Moreover, the stacked structure model outperforms the parallel model, with the former exhibiting ~ 1 % lower MAPE, thereby demonstrating that the stacked structure can more effectively maintain the integrity of spatial dependencies. The accurate metro passenger flow prediction achieved in this study has significant implications for guiding both the daily metro train scheduling to better meet real-time passenger requirements (Yin et al., 2017) and the accurate passenger flow control to ensure safety during peak periods (Shi et al., 2019).

In addition to the three spatial graph elements we have chosen (station adjacency graph, passenger flow similarity graph, and passenger flow correlation graph), other spatial feature graphs effectively represent the spatial dependencies between metro stations. These include the travel distance graph and population flow graph, as supported by the First Law of Geography and previous studies (Ma et al., 2018; An et al., 2019; Chen et al., 2021). For future research, it is recommended to explore additional spatial feature graphs or construct new graph elements based on prior knowledge to enhance prediction accuracy. While we did not incorporate external factors to predict metro passenger flow due to limited external data resources, it is worth noting that factors such as holidays and weather types have been found to influence travel demand in previous studies (Shen et al., 2017; Liu et al., 2021). Therefore, integrating these factors into the model could provide further insights into the dynamic evolution patterns of passenger flow and improve prediction accuracy. Currently, scholars have started conducting metro passenger flow forecasting under abnormal scenarios (Xue et al., 2022; Zhang et al., 2022). It is recommended that future research also considers forecasting metro passenger flow under abnormal scenarios. As for future models, incorporating attention-based modules could be considered to enhance performance and foster innovation.

CRedit authorship contribution statement

Shuguang Zhan: Data curation, Methodology, Writing – original draft, Conceptualization, Software, Validation. **Yi Cai:** Investigation, Writing – review & editing, Validation. **Cong Xiu:** Writing – review & editing, Methodology, Supervision. **Dajie Zuo:** Writing – review & editing, Methodology, Supervision. **Dian Wang:** Writing – review & editing, Methodology, Supervision. **Sze Chun Wong:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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